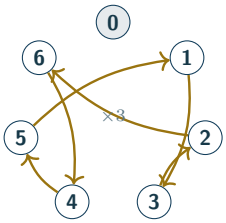
NUMBER THEORY LADDER 3 OF 9

Arithmetic on a clock: congruences and inverses

Two numbers are congruent when they differ by a multiple of n, and sums and products respect that.

1 / THE RULE AND THE PICTURE

Working modulo 7 keeps only the remainder, and multiplying by 3 walks round the whole clock



We write $a \equiv b \pmod{n}$ when $n \mid a-b$, for example $9 \equiv 2 \pmod{7}$, and congruences can be added and multiplied:

$$a \equiv a',\ b \equiv b' \implies a+b \equiv a'+b',\quad ab \equiv a'b'$$

Starting at 1 and multiplying by 3 repeatedly visits $1, 3, 2, 6, 4, 5$ and returns to 1, using every non-zero remainder, so 3 generates the clock.

2 / MULTIPLICATION TABLES / A PRIME CLOCK AND A COMPOSITE ONE

Modulo a prime every row is a shuffle of the clock, while $2 \cdot 3 \equiv 0 \pmod{6}$ breaks that

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

modulo 7, with the 1s in gold

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	0	2	4
3	3	0	3	0	3
4	4	2	0	4	2
5	5	4	3	2	1

modulo 6, with the zeros in red

In the table for 7 each non-zero number meets 1 exactly once, so it has an inverse: $2 \cdot 4 \equiv 1$, $3 \cdot 5 \equiv 1$, $6 \cdot 6 \equiv 1$. In the table for 6 the numbers $2, 3, 4$ make zeros and 5 alone has an inverse besides 1.

3 / WHEN AN INVERSE EXISTS / UNITS

a has an inverse modulo n exactly when $\mathrm{HCF}(a, n) = 1$

n	units	$\varphi(n)$	inverses
7	$1, 2, 3, 4, 5, 6$	6	all non-zero, a field
12	$1, 5, 7, 11$	4	each is its own inverse
10	$1, 3, 7, 9$	4	$3 \cdot 7 \equiv 1$, $9 \cdot 9 \equiv 1$
15	$1, 2, 4, 7, 8, 11, 13, 14$	8	$2 \cdot 8 \equiv 1$, $4 \cdot 4 \equiv 1$

If $\mathrm{HCF}(a, n) = 1$ then $ax + ny = 1$ by Bézout, so $ax \equiv 1$ and x is the inverse, and if the HCF is larger then every product ax shares that factor with n and can never be 1.
The number of units is Euler's $\varphi(n)$, and modulo 12 we have $5^2 = 25$, $7^2 = 49$, $11^2 = 121$, all $\equiv 1$.

4 / USING CLOCKS TO CHECK ARITHMETIC / CASTING OUT NINES AND ELEVENS

A product must survive reduction modulo 9 and modulo 11

Since $10 \equiv 1 \pmod{9}$, a number is congruent to its digit sum, and since $10 \equiv -1 \pmod{11}$ to its alternating digit sum.
For $1234 \times 5678 = 7\,006\,652$ the digit sums are $1 \times 8 = 8$ on the left and $7+0+0+6+6+5+2 = 26 \to 8$ on the right, and the alternating sums from the right are $2 \times 2 = 4$ and $2-5+6-6+0-0+7 = 4$, so both clocks agree. A wrong answer would have to be wrong by a multiple of 99 to survive both checks.

ASIDE / ARITHMETIC WITHOUT LARGE NUMBERS

Powers reduce at every step: $3^{100} \bmod 7$ needs only the cycle of length 6, and $100 \equiv 4$, so it is $3^4 = 81 \equiv 4$.

LINKS TO OTHER SHEETS

Units and inverses use *Euclid's algorithm and Bézout's identity*; the next sheet is *Fermat, Euler and repeating decimals*.

Convention. $\equiv$ is congruence, $\varphi(n)$ the number of units modulo n; tables list products reduced modulo n.